Mohammed Shakib Hossain

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Research Interests

Usable Security and Privacy, Human-Computer Interaction (HCI), Human-Centered Security and Privacy, Empirical and Behavioral Security, Human Factors in Security, Human-AI Interaction, Human-Centered AI, Privacy and Trust in AI Systems, Security and Privacy in Emerging Technologies, Authentication and Security Protocols.

Education

BRAC University — Dhaka, Bangladesh

Master of Science in Computer Science and Engineering — *Ongoing*

- **Current CGPA: 4.00/4.00**
  Relevant Coursework: Introduction to Cryptography, Applied Cryptography, HCI and Design, Information Assurance and Human Factor, Usable Security and Privacy, Computer Communications and Networks, Blockchain and Distributed Ledger, Research Methodology and Seminar

BRAC University — Dhaka, Bangladesh

Bachelor of Science in Computer Science and Engineering — *June 2021 – Jan 2025*

- **CGPA: 3.93/4.00** — ***Highest Distinction***
  Academic Highlights: Consistently Earned A/A+ grades in core CSE and mathematics courses, including Programming Languages, Data Structures and Algorithms, Operating Systems, Computer Architecture, Theory of Computation, Compiler Design, Blockchain and Cryptocurrency, Artificial Intelligence, and Advanced Mathematics.

Research Experience

MSc Thesis Research — 2026 – Present

Department of Computer Science and Engineering, BRAC University — *Dhaka, Bangladesh*

- Exploring why employees deviate from organizational security policies when formal requirements conflict with workplace demands, practical constraints, or established work practices.
- Conducted preliminary exploratory interviews with software/IT professionals and reviewed literature on security policy compliance, human factors, and usable security.
- Investigating how employees, security managers, and other organizational decision-makers interpret, respond to, and manage policy deviations and exceptions in everyday work.

BSc Thesis Research — 2024 – 2025

Department of Computer Science and Engineering, BRAC University — *Dhaka, Bangladesh*

- Designed an NFT-based authentication mechanism for federated metaverses, addressing usability challenges in cross-platform authentication.
- Developed a proof of concept integrating MetaMask wallet authentication with a VR metaverse and a Minecraft Paper server, using Next.js and Node.js.
- Developed a threat model and formally verified the proposed authentication protocols using ProVerif.
- Evaluated usability using the System Usability Scale (SUS) and a custom UX questionnaire informed by ISO 9241-11 and Nielsen's usability heuristics.
- Thesis repository: [BRAC University Repository]

Publications

Conference Papers

- **M. S. Hossain**, K. E. U. Hawak, M. Jawad, M. Y. Ali, S. Karmoker, M. A. Nahid, H. Shahriar, and M. S. Ferdous, *"A Secure Authentication Mechanism for Federated Metaverses Using Non-Fungible Tokens (NFTs)"*, *IEEE International Conference on Blockchain (Blockchain 2026)*, IEEE Cybermatics Congress, Montbéliard, France, 2026. **Accepted for publication** [Conference]

Manuscripts in Preparation

- **M. S. Hossain**, *"Negotiating Security in Everyday Organizational Work: How Security Policy Deviations and Exceptions Are Interpreted and Managed in Bangladesh,"* manuscript in preparation, 2026.

Research Projects

Before You Share: Designing a User-Controlled Privacy Review Interface for Everyday AI Advice-Seeking Among Bangladeshi University Students

CSE731: HCI and Design Research *Fall 2026*

- Proposed a study of how university students balance contextual relevance and personal privacy when seeking everyday advice from AI chatbots.
- Designed a mixed-method research plan and clickable Figma prototype enabling users to retain, generalize, or remove personal details while preserving useful context.

Understanding Security Policy Violations and Exceptions in Organizational Workflow in Bangladesh

CSE791: Research Methodology and Seminar *Summer 2026*

- Investigated information-sharing and client-communication practices among software/IT employees in Bangladesh, focusing on workplace demands, communication-channel choices, informal security expectations, and client requirements.
- Analyzed five semi-structured interviews using reflexive thematic analysis, identifying four themes; contributed to research framing, literature review, participant recruitment, interview coordination, data analysis, thematic interpretation, and manuscript preparation.

Bridging the Gap: Understanding Security Policy Violations in Organizational Workflows in Bangladesh

CSE726: Information Assurance and Human Factors *Spring 2026*

- Proposed a human-factors study examining tensions between formal security policies, workplace demands, and practical constraints.
- Prepared the introduction, literature review, study design, and methodology, focusing on information assurance and organizational security practices.

Living Under Watch: Privacy, Surveillance, and Digital Harm among Tea Garden Workers in Bangladesh

CSE773: Usable Security and Privacy *Spring 2026*

- Proposed a qualitative study of privacy, surveillance, and digital harm among tea garden workers, considering how social and demographic differences shape their experiences and risks.
- Prepared the introduction, literature review, study design, and methodology to inform inclusive, human-centered security and privacy approaches.

Research Methods

Qualitative Research: Semi-Structured Interviews, Reflexive Thematic Analysis (Braun & Clarke)
User Research & Usability Evaluation: User Studies, System Usability Scale (SUS), Likert-Scale UX Surveys
Quantitative Analysis: Descriptive Statistics, Likert-Scale Response Analysis

Work Experience

Undergraduate Teaching Assistant (Student Tutor) Oct 2023 – May 2025

Department of Computer Science and Engineering, BRAC University *Dhaka, Bangladesh*

Courses taught: Algorithms (CSE221), Discrete Mathematics (CSE230)

- Conducted tutorials and individual consultations, supporting 100+ students in Discrete Mathematics, Algorithms, and problem-solving.
- Prepared problem sets and assessment rubrics aligned with course learning outcomes to reinforce conceptual understanding.
- Evaluated assignments and quizzes and provided feedback to support students learning and performance.
- Collaborated with course instructors to facilitate tutorials and hands-on activities and provide technical guidance to students.

Honors & Awards

100% Merit-Based Graduate Scholarship — Dhaka, Bangladesh
Full tuition waiver for outstanding academic merit and performance in the MSc program. — *2026 – Present*

100% BRAC Scholarship — Dhaka, Bangladesh
Awarded full tuition support for outstanding academic performance in undergraduate studies. — *June 2021 – Jan 2025*

Vice Chancellor's List (9 of 11 Semesters) — Dhaka, Bangladesh
Recognized for achieving a GPA between 3.90 and 4.00 in individual semesters. — *June 2021 – Jan 2025*

Graduated with Highest Distinction — Dhaka, Bangladesh
Conferred upon completion of the BSc with a CGPA of 3.80 or above. — *Jan 2025*

Technical Projects (Selected)

Chat Secretly – Cryptography-Based Secure Messaging System | *FastAPI, Python, SQLite* — 2026

- Collaborated on a secure messaging platform with user authentication, using FastAPI and SQLite.
- Implemented and analyzed AES-128, DES, RSA, ECC, substitution, and double transposition ciphers; developed a Crypto Lab for encryption/decryption, key generation, and cryptographic attack demonstrations.
- Project source: [GitHub Link]

StreamPay – Decentralized Payroll Streaming Protocol | *Solidity, Foundry, Ethers.js v6* — 2026

- Built a decentralized payroll protocol that locks ETH for employees and streams funds continuously on a per-second basis until withdrawal.
- Developed and tested Solidity smart contracts with Foundry and built an HTML/CSS/JavaScript frontend using Ethers.js v6 to interact with the contracts.
- Project source: [GitHub Link]

Transformer-based PDF Summarization System | *BERT, BART, NLP* — 2026

- Developed a PDF summarization pipeline using text extraction, preprocessing, and BART-based summarization, with chunking and a map-reduce strategy for long documents.
- Evaluated summary quality using ROUGE-1, ROUGE-2, and ROUGE-L metrics.
- Project source: [GitHub Link]

Dkash – Online Transaction Platform | *MongoDB, Express.js, React, Node.js* — Fall 2023

- Collaborated on a digital transaction platform using the MERN stack, with an emphasis on secure transaction workflows.
- Applied software engineering practices, including requirements analysis, use-case modeling, system architecture, and interface design.
- Project source: [GitHub Link]

Technical Skills

Languages: Python, C/C++, JavaScript, Solidity
Frameworks & Libraries: React, Node.js, pandas, NumPy, Matplotlib, Pytorch
Developer Tools: VS Code, PyCharm, Code::Blocks, LTspice, emu8086
Core Competencies: Problem-solving, Team Collaboration, System Design

Languages

- **English:** Professional working proficiency
- **Bangla:** Native proficiency

References

Dr. Md Sadek Ferdous — Professor
Department of Computer Science and Engineering, BRAC University — *Email: sadek.ferdous@bracu.ac.bd*

Dr. S M Taiabul Haque — Professor
Department of Computer Science and Engineering, BRAC University — *Email: taiabul.haque@bracu.ac.bd*